

is used to deliver sensor data between components to support low-latency processing and control applications," explains the paper by Michael Blackstock and Rodger Lea from Media and Graphics Interdisciplinary Centre at the University of British Columbia, Vancouver, Canada.

The above concept actually builds on previous IoT research. This toolkit approach, embodied by WoTKit, has been commercialised by Sense Tecnic Systems.

There are various operating systems that have been developed keeping memory-constrained devices in mind. One such example is Contiki. It is an Open Source operating system for networked, memory-constrained systems with focus on low-power wireless IoT devices. It has been used in street-lighting systems, sound monitoring for smart cities, radiation monitoring systems and alarm systems.

Contiki was created by Adam Dunkels in 2003, and has since been further developed by developers from Atmel, Cisco, Oxford University, SAP, STMicroelectronics and others. This operating system supports recent low-power wireless standards like 6LoWPAN, RPL and CoAP.

Applications are developed on Contiki in Standard C programming language. With Cooja simulator Contiki networks can be emulated before these are burnt into the hardware, thus reducing the time required to develop and saving you the cost of prototyping every design. Instant Contiki also provides an entire de-

Important milestones that engineers should keep track of

1. European Commission consults on rules for wirelessly connected devices

Nellie Kroes, European Commission vice-president for Digital Agenda, said: "An Internet of Things with intelligence embedded into everyday objects is the next big thing. I want to promote an Internet of Things that serves our economic and societal goals, whilst preserving security, privacy and the respect of ethical values."

Realising the enormous economic and societal potential of the IoT requires a level-playing field where all players can compete on an equal footing, without gate keepers and locked-in users. Its societal acceptance requires the definition of an ethical and legal framework, supported by technology and providing people with control and security. The results of the consultation will feed into the Commission's Recommendation on the IoT.

2. Special Interest Group signs agreement to standardise the Internet of Things

ARM, Cable & Wireless Worldwide, CSR and Neul have formally signed the SIG Promoter Agreement that details how they will back the mission to establish a new standard and encourage its adoption worldwide. The Weightless standard that this group will define is on track for completion in early 2013.

"The Internet of Things requires new thinking about technology. As data levels soar across the world, new ways need to be found to ensure wireless communication can be seamless. This includes the next wave of connectivity across smart grids, enhanced healthcare, smart cities, asset tracking, sensors and future applications as yet unimagined. With common standards we can all benefit from intelligence embedded and connected everywhere, so the ARM team is excited about the huge potential this standard will unlock," said Mike Muller, CTO, ARM.

velopment environment in a single download.

The next wave of connectivity

The number of devices connected to the Internet is growing so fast that it has already outnumbered the world population. This rapid growth is what prompted the switch to IPv6 last year so that we do not run out of IP addresses. Unlike IPv4, IPv6 allows us to have around 3.4×10^{38} IP addresses. (For more about IPv6, read the author's article "Are you ready to transit to IPv6?" published in EFY March 2012 issue.)

The next wave of 'getting connected' focuses on devices that so far

remained immune to getting connected, such as fridges, washing machines and other home equipment. "In this Internet of Things, we are trying to give them sensors so that they can be more intelligent in order to be aware of their context and thus operate with a much better performance," explained Janardan Revuru in his talk 'Internet of things—The Rise of Unintelligent Machines' at Electronics Rocks 2012.

Some of the technical challenges in the IoT include communication challenges with respect to low speed and high error rates combined with power considerations and small form factor required. The devices also feature an extremely small memory footprint, which means that the software on-board would be very restricted in terms of size.

In order to optimise power consumption and communication performance, these devices use some low-power communication standards. Some of the important standards related to Internet of Things are IEEE 802.15.4, 6LoWPAN, RPL and CoAP.

The IEEE 802.15.4 standard offers physical and media-access control layers for low-cost, low-speed and low-power, wireless personal-area



Unlike the telecom revolution, where a new device came into existence either as a phone on desk or as mobile in hand, the IoT revolution will be hard to accredit. The IoT will happen by raising the bar for intelligence of things already existing around us. The next model of watch could be IoT-ready and you won't physically see it different. It will measure your heart beat, blood pressure and metabolic parameters to assess your health. It is your personal doctor!" — Janardan Revuru, project manager, HP